EIGSEP Memo 004

Marjum Pass 2026-07: receiver and temperature calibration

EIGSEP Team

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Abstract

How box-air integrations are put on a kelvin scale, what that scale means, and how far to trust it. The calibration is the minimal three-state Y factor with reflection corrections, built for every integration from the noise-source, load and S11 measurements nearest in time and the lab characterization of the RF-switch paths. It is published as the data product `tcal` and applied by `eigsep_data` when data are loaded. This memo states the method, its validation and coverage, the contaminated calibration visits it drops, where the scale fails at low frequency, and what remains open, above all the noise-source temperature. Every figure and number is generated by `make_products.py` in this directory (Appendix A). The measurement-model derivation is `eigsep_cal` memo M003; the dataset is memo 001.

1 Summary

- **What a calibrated temperature is.** The source temperature of the antenna path at reference plane P, the LNA-side port of the RF switch. It includes the antenna, balun, balun–switch coax and switch path; none is removed. It is not a free-space antenna temperature (§2).
- **The product.** `tcal` v0003 (`marjum-2026-07/derived/tcal/v0003/`); `Bundle.calibrated` in `eigsep_data`. It covers 63 % of box-air antenna integrations between 07-17 04:11 and 07-18 02:58 UTC (73,885 of 116,640); everything else is NaN, never extrapolated (§5).
- **It agrees with an independent calibration** of the 16/17 July night (Christian's, 2026-09-25) to a median ratio of 1.0003 per sky visit (§5), and with Christian's S11 chain to 1×10^{-15} in Γ at P (§4.3).
- **Three calibration visits are dropped.** They carry box-air's 1.000 MHz comb EMI for a minute or two (§4.2).
- **The scale fails below about 30 MHz** and is untrustworthy to about 60 MHz, where the reflection corrections are large and treat loss in the balun and coax as mismatch (§6).
- **Open: the noise-source temperature.** The scale rests on the nameplate 917 K. The one in-band route to it gives a value 2.4 times lower and cannot be trusted either (§7). Against a GSM × HFSS prediction the data sit 1.61–1.29 times high at 78–150 MHz, which a lower T_{NS} would reduce.

2 Reference plane P and the switch network

The RF switch connects the LNA to one of several sources; plane P is its LNA-side common port (`eigsep_cal/docs/api.md` §3). Each switch state's path is part of its source. Table 1 lists the states used here.

Calibration is done at P and nothing is de-embedded from a temperature (decision recorded in api.md §5.2). A comparison with a free-space simulation therefore has to carry the simulation forward through the balun, coax and switch path, not the data back (§6).

state	source behind the switch path	role
`RFANT`	antenna, balun, balun–switch coax	the sky
`RFAMB`	50 Ω ambient load (thermistor `tempctrl_load`)	load reference
`RFNON`	noise diode on, behind a 30 dB pad	noise-source reference
receiver	LNA input, measured through the VNA leg	$\Gamma_{\rm rec}$

Table 1: Switch states. Only box-air (correlator input 4 in phase C) is switched; box-gnd cannot be calibrated this way.

3 Measurement model

The three-state Y factor is

$$T^* = T_{\rm load} + T_{\rm NS}\,\frac{P_{\rm ant} - P_{\rm load}}{P_{\rm NS} - P_{\rm load}}, \tag{1}$$

with $T_{\rm load}$ the load's physical temperature and $T_{\rm NS}$ the noise-source excess over it. With the mismatch factor $M_s = (1-|\Gamma_s|^2)/|1-\Gamma_s\Gamma_{\rm rec}|^2$ and powers $P_{\rm ant} = g(M_{\rm ant}T_{\rm ant} + T_r)$, $P_{\rm load} = g(M_{\rm load}T_{\rm load} + T_r)$, $P_{\rm NS} - P_{\rm load} = gT_{\rm NS}$,

$$T_{\rm ant} = \frac{T^* - (1 - M_{\rm load})\,T_{\rm load}}{M_{\rm ant}}. \tag{2}$$

This is the minimal tier of M003 §7.1. It assumes equal gain on every switch path, no receiver noise waves, and that $T_{\rm NS}$ is the excess the receiver takes. Both steps are linear in the measured power, so a calibration is a pair of spectra, $T = s\,P + o$ (`eigsep_cal.dicke.tstar_coefficients`, `receiver_s11_coefficients`), that can be applied to any integration.

4 Inputs

4.1 Calibration visits

From 07-17 04:10 UTC the switch visits RFNON and RFAMB about every 13 minutes. Each visit becomes one spectrum: an integration-time-weighted mean over its samples. Samples are masked by the RFI bits of the campaign flag product `flags@v3` (memo 006; released unchanged from `v3-beta.3`), which flags calibration rows against their own backgrounds, and, directly, when wrapped (int32 overflow) or in a dropped integration, which the flag product does not cover as RFI. At most 29 % of a visit's samples are masked. Two clock-aligned channels (125.0 and 187.5 MHz) are fully flagged in a few visits, and the calibration is NaN there for the integrations those visits bracket. 67 load and 65 noise-source visits are kept. Each carries the receiver regime of its window from `curation/cal_windows.jsonl`, matched by file, because the window bounds are file close times.

Table 2 summarizes the 16/17 July night (21:30–07:55 MDT), the data set behind the spectra and comparisons below.

	16/17 July night
box-air integrations (input 4)	69849
correlator files	292
integrations timed from the filename (stale clock)	0
dropped (all-zero) integrations	0
integrations with an int32 wrap	289
channels with an int32 wrap	16
integrations without a load-thermistor reading	190
visits: antenna / noise source / load	54/48/49
longest time from a sky visit to the nearest load visit (min)	26.5
longest time from a sky visit to the nearest noise-source visit (min)	25.6
load temperature range (K)	298.46–301.53

Table 2: The 16/17 July night, box-air: what the calibration had to work with.

4.2 Contaminated visits

Three visits disagree with their neighbours in 278–296 channels by more than 10 %; no other rx-A visit exceeds 4 such channels, and rx-B's drifting edge visits reach 48. They are dropped when more than 100 channels disagree. Figure 1 shows what is in them: a comb of narrow lines at integer MHz, the odd-MHz teeth strongest. Fourier transformed across channels after removing a running median, each visit has comb power 6.6×10^5–8.0×10^5 times the noise at 1.000 MHz (4.096 channels) and at most 527 times at 1.953125 MHz (8 channels, the transmitter grid); their clean neighbouring visits show at most 2 times at either. This is the signature of box-air's 1.000 MHz comb EMI of 07-16 (memo 001), and Christian's notebook calls the 08:53 visit "the 1 MHz comb burst". Per integration it occupies the first minute or two of each visit and is absent on box-gnd (memo 006, § "The self-EMI in calibration visits on 07-17", which gives it per integration and masks it as `boxair-emi-1mhz`). It is not the transmitter, whose 8-channel comb first appears at 15:36:22 UTC. The identification rests on the spacing and on which antenna sees it, not on a physical test. All three visits also share a calibration window with an S11 sweep; whether that is related, and what switches the comb EMI on at the start of a calibration visit, are open (memo 006).

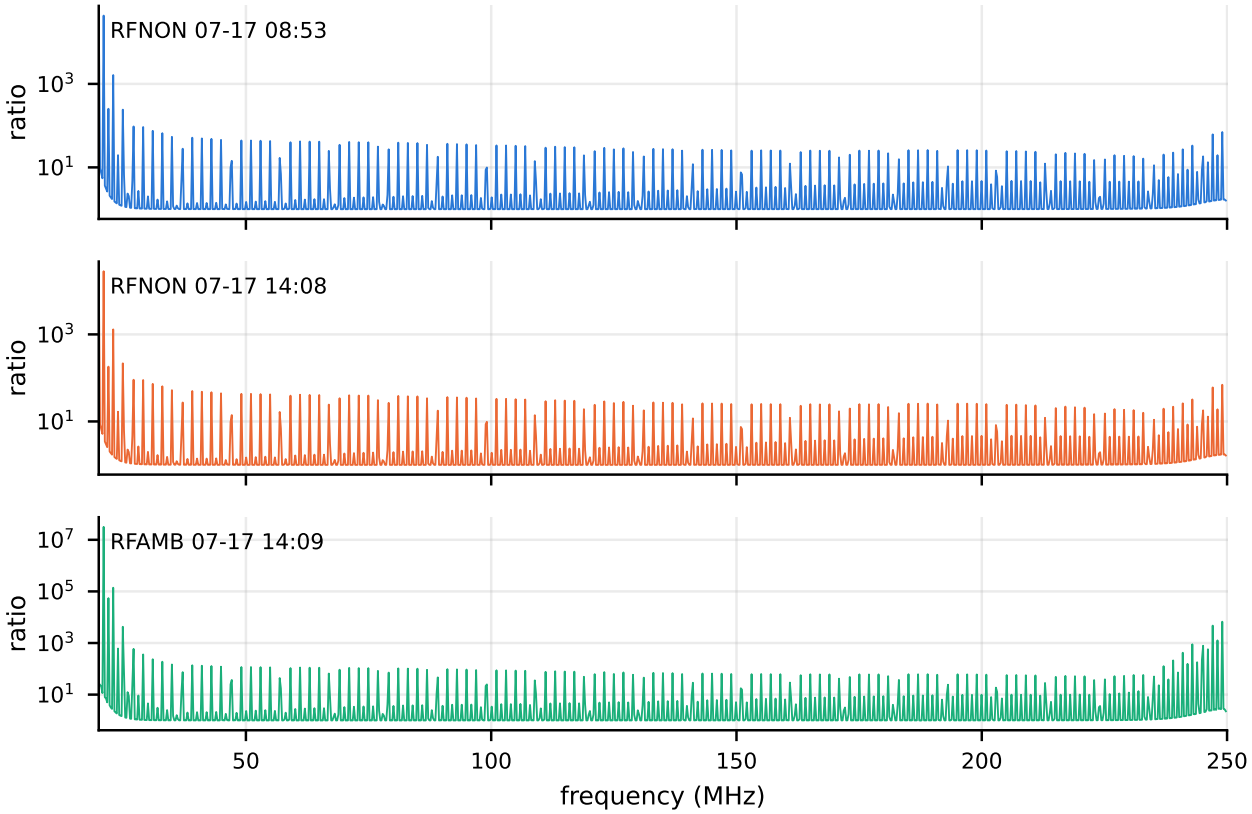


Figure 1: The three dropped calibration visits (raw integrations, weighted mean) divided by the mean of the two clean visits on either side (tcal v0003).

4.3 S11 at plane P

40 raw VNA sweeps (`data/s11/`) are calibrated with `eigsep_cal.S11`: the VNA error network is solved from each sweep's internal open/short/load against their lab characterization, then the VNA-leg switch path is de-embedded and the RF-leg path embedded. A sweep whose own internal standards are zero-filled borrows the nearest valid set. This agrees with Christian's independent chain to 1×10^{-15}, and `eigsep_cal`'s packaged internal-standard characterization equals the one derived from the raw lab capture to 9×10^{-16}.

The switch-path S-parameters are Charlie's characterization (`data/s11/cal_materials/switch_sparams.npz`,

packaged in `eigsep_cal` since 96d4e9e). It solves the VNA's error network from its standards first and then characterizes each switch path against standards placed at its port.

4.4 Load temperature and noise-source excess

$T_{\rm load}$ is the `tempctrl_load` thermistor reading nearest each integration (logged whether or not the switch cycles; 300.0 K on the night). $T_{\rm NS} = 290\,{\rm K} \times 10^{0.5} = 917\,{\rm K}$ is the nameplate excess, 35 dB ENR behind a 30 dB pad. It has not been measured (§7).

5 The tcal product

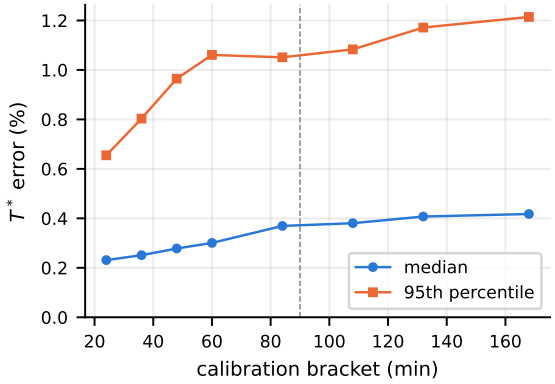
5.1 How each integration is calibrated

For each integration, $P_{\rm load}$ and $P_{\rm NS}$ are interpolated linearly in time between the visits that bracket it, within one receiver regime and only across a gap of at most 90 min. Each Γ is interpolated between the bracketing sweeps of its epoch (gap at most 3 h), or held from the nearest sweep within 2 h; sweeps are never used across the 07-17 19:11 → 07-18 01:17 gap, which spans the regime boundary. $T_{\rm load}$ is the nearest thermistor reading within 60 s. Anything outside a limit is NaN. The calibrated view is NaN on rows that are not `RFANT`; `Bundle.t_star` gives T^* for any state.

5.2 Why these limits

Leave-one-out tests (`validation.json`) predict each interior visit from its neighbours. For the gain proxy $P_{\rm NS} - P_{\rm load}$ the median error is 0.21 % linear and 0.33 % for the nearer neighbour; linear wins in 71 % of cases. Widening the bracket (Figure 2, Table 2) costs little: the error levels off near 0.4 % median beyond about 90 minutes. Holding an S11 sweep for 1 or 2 h moves the calibrated temperature by 0.03 and 0.04 % (95th percentile at most 0.13 %).

The 177-minute gap in rx-A (07-17 16:22 → 19:19) is not bridged. The gain proxy changes by 8.4 % across it, against a median of 2.4 % and a maximum of 5.3 % over the same span elsewhere; the antenna was raised to 91 m inside it. The antenna S11 changes by 2.1×10^{-3} across it, inside its usual range (95th percentile 3.5×10^{-3}).



bracket (min)	median (%)	95th pct (%)	max (%)
24.0	0.23	0.65	1.1
36.0	0.25	0.80	1.5
48.0	0.28	0.96	1.8
60.0	0.30	1.06	2.4
84.0	0.37	1.05	2.5
108.0	0.38	1.08	2.5
132.0	0.41	1.17	2.7
168.0	0.42	1.21	2.7

Figure 2: Error in T^* from interpolating the calibration across a wider bracket, on rx-A (50–200 MHz, on the night's mean sky). The dashed line is the 90-minute limit.

5.3 Coverage, drift and validation

Figure 3 shows where the product is defined. Its lower panel reproduces the noise-source drift of legacy MEMO-004 independently: the injected excess relative to the load power, which is immune to receiver gain, rises 16.3 % peak to peak over the 12.2 h before the hole (correlation with time +0.90), while the load power holds to 0.85 %

rms. Across the hole it steps up further. Interpolating between cycles follows this drift; nearest-visit pairing does not.

Through `load_bundle`, T^* on load rows returns the thermistor temperature to a median of 0.006 K (95th percentile 1.2 K) and on noise-source rows $T_{\rm load}+T_{\rm NS}$ to 0.29 K (8.6 K). The worst residuals are the contaminated rows themselves. Against Christian's calibration, per sky visit over 50–200 MHz, the ratio has a median of 1.0003 (5th–95th percentile 0.9983–1.0017); both sides take per-visit medians there, because Christian's chain does.

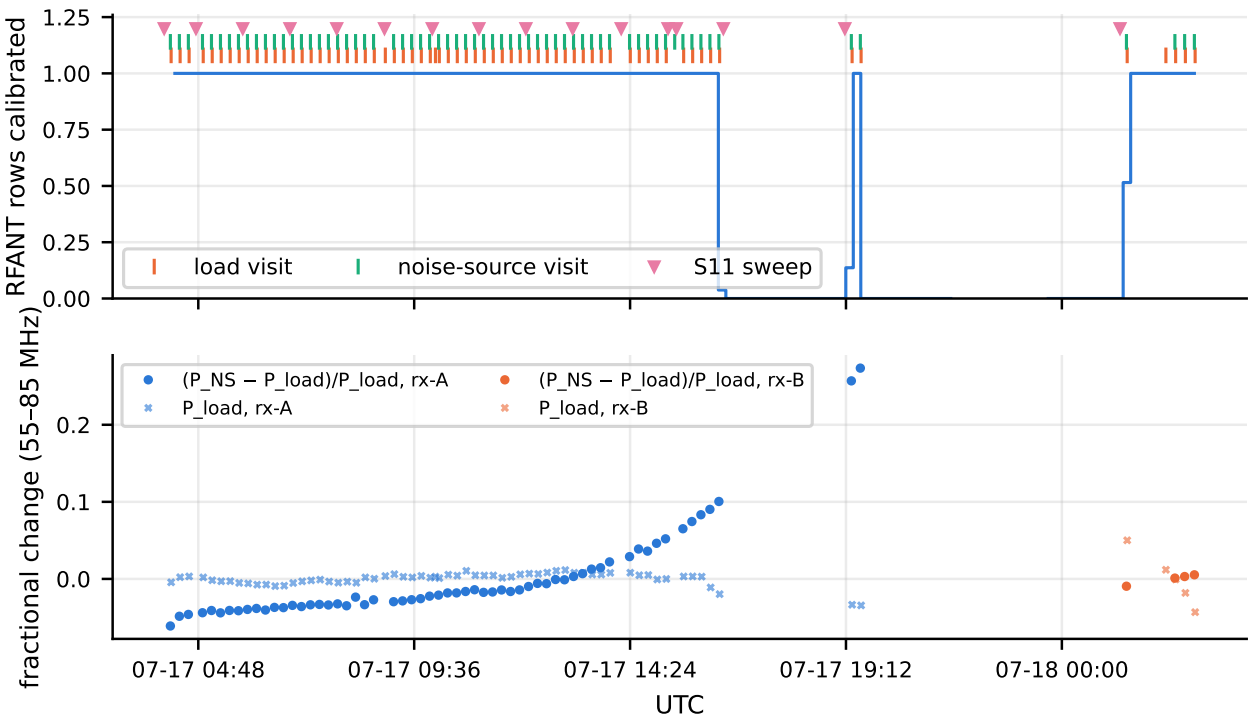


Figure 3: Top: fraction of box-air antenna integrations calibrated, per 10 minutes, with the calibration visits and S11 sweeps. Bottom: noise-source excess over load power and load power, 55–85 MHz, relative to their means.

6 The spectrum, and where the scale fails

Figure 4 shows the night's calibrated spectrum: weighted means over antenna integrations, with the `flags@v3` RFI bits and wrapped samples as zero weight. The RFI bits mask 8.8 % of antenna samples on the night and 68 % in the FM band. They do not cover int32 wraps: 227 wrapped samples, each near -2×10^9 counts, would otherwise put deep false dips in the aircraft band. A channel with less than half its integration time left after masking is dropped (74 channels, most of the FM band), because the RFI bits flag mostly positive excursions and the remainder is biased low.

Below about 30 MHz the scale has failed. The receiver passes almost nothing there: the load power is zero counts at 5–15 MHz (Table 3). When the antenna and load powers round to the same count, T^* returns $T_{\rm load}$; where the noise-source and load powers are equal it divides by zero. The values near 300 K are the formula, not the sky.

From 30 to about 60 MHz the result rests on the reflection corrections. The antenna reflects most of its power at P ($|\Gamma_{\rm ant}|\approx 0.7$–$0.8$), $|\Gamma_{\rm rec}|$ peaks near 0.7, so the reflection corrections are large there ($M_{\rm ant}$ in Table 3). The bump in T^* near 36 MHz is removed by the receiver term ($M_{\rm ant}>1$ there). The correction treats every reduction of $|\Gamma|$ at P as mismatch; but the balun, coax and switch path between the antenna and P

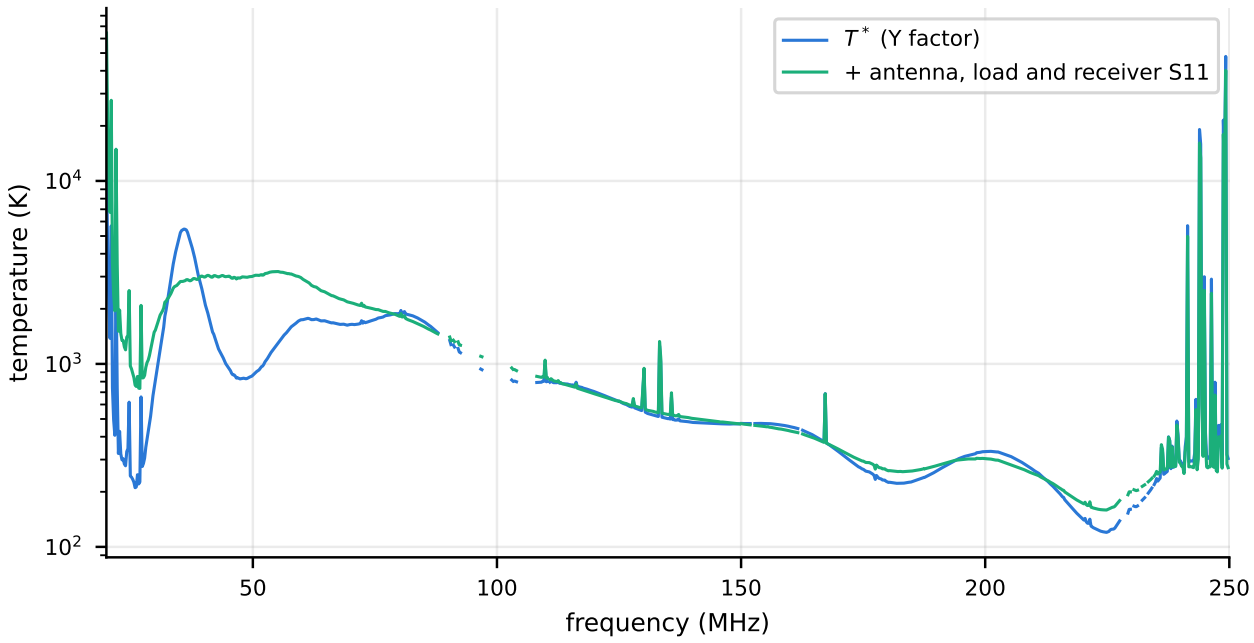


Figure 4: The 16/17 July night through tcal v0003: T^* and the fully corrected temperature, weighted means over antenna integrations.

absorb power and re-emit at their physical temperature, which the S11 cannot see. That loss is largest where the antenna is worst matched. Treat the band below about 60 MHz as uncalibrated until the balun is characterized.

MHz	P_{load} (counts)	$\lvert\Gamma_{\mathrm{ant}}\rvert$	M_{ant}	T^* (K)	T (K)
4.9	9.85	0.83	0.50	879	1801
10.0	2.38	0.81	0.29	880	3078
20.0	2.45	0.74	0.20	13081	64304
24.9	701	0.72	0.25	244	974
30.0	2.8e+04	0.71	0.52	799	1530
34.9	9.62e+04	0.72	1.78	4988	2790
40.0	1.98e+05	0.74	0.74	2223	2982
50.0	4.42e+05	0.74	0.28	870	3023
60.1	6.04e+05	0.62	0.59	1741	2923

Table 3: The low band on the night: mean load power, antenna reflection and mismatch at P (night-mean sweep), and the weighted-mean temperatures.

Against a sky model. Figure 5 compares the calibrated spectrum with GSM2016 seen through the HFSS beam normalized to directivity, pointing at zenith as it did that night, with 300 K ground filling the lower hemisphere (the bowtie radiates 0.41–0.63 of its power upward). The simulation rises monotonically; the data flatten below 60 MHz and sit high above it: data/simulation is 0.88 at 51 MHz, 1.42 at 70, 1.61 at 78, 1.34 at 129 and 1.29 at 148 MHz. The simulation is at the antenna terminals and the data at P, so the comparison cannot separate the candidates: a lower T_{NS} scales the data down, while balun and coax loss and noise waves change its shape.

The HFSS "efficiency" is port mismatch. The HFSS gain integrates to well under 4π at low frequency. Dividing by the mismatch of HFSS's own input impedance against a 75 Ω port leaves a radiation efficiency near 1 from 51 to 100 MHz (Table 4); against 50 Ω it is not flat. So the HFSS port was 75 Ω, the simulated radiator has negligible ohmic loss, and the HFSS gain must not be applied on top of the measured mismatch correction.

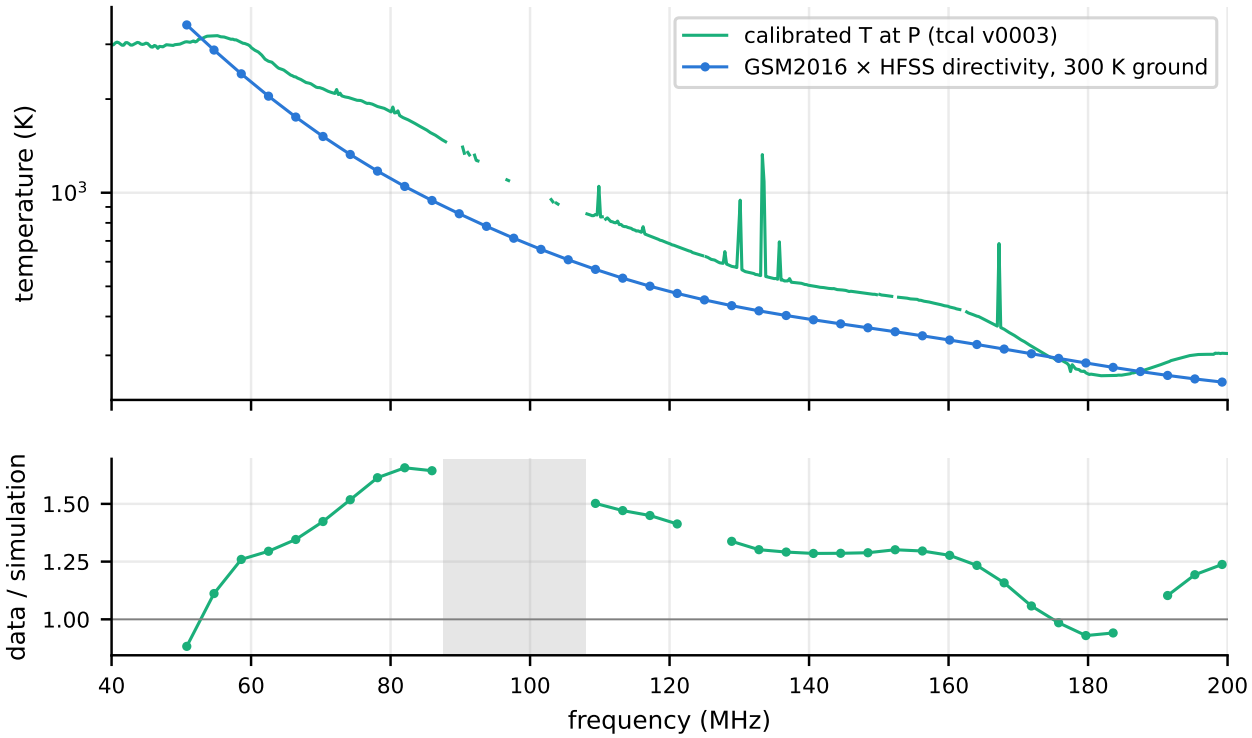


Figure 5: Calibrated temperature on the night against GSM2016 × HFSS directivity, weighted by the calibrated integrations in each 15-minute bin. The FM band is left out of the ratio.

MHz	Z_{HFSS} (Ω)	$\int G\,d\Omega/4\pi$	$1-\lvert\Gamma_{75}\rvert^2$	e_{rad} (75 Ω)	e_{rad} (50 Ω)
51	$18.5-61.5j$	0.453	0.444	1.02	1.04
59	$21.3-42.5j$	0.562	0.576	0.98	0.91
70	$26.6-20.8j$	0.702	0.742	0.95	0.83
78	$31.9-7.1j$	0.791	0.834	0.95	0.84
86	$39.0+5.3j$	0.865	0.898	0.96	0.88
100	$56.9+23.0j$	0.959	0.952	1.01	1.01

Table 4: HFSS realized gain split into port mismatch and radiation efficiency (`eigsep_sim` bowtie v000 impedance; the beam integral is `bowtie_beam.npz`).

7 Open questions

- **The noise-source temperature.** Legacy MEMO-004 computed T_{NS} in-band from the single noise-diode-off visit (07-16), treating the load and the diode-off path as passive loads at their thermistor temperatures. It gives a value 2.4 times below nameplate, but the same premise fails a closure test by the same factor, so it is not a measurement: either the nameplate is 2.4 times high, or the two paths differ in loss or mismatch. The noise-source path's switch S-parameters (VNANON, RFNON) were never measured, and with $|\Gamma_{\mathrm{rec}}|$ up to 0.7 its mismatch can differ strongly from the load's. A bench measurement of the diode and its path settles it; it is the largest uncertainty in the scale.
- **Receiver noise waves.** Omitted from `tcal`. With $|\Gamma_{\mathrm{ant}}|$ and $|\Gamma_{\mathrm{rec}}|$ both large below 60 MHz they cannot be negligible there. A first tier-2 fit (M003 §7.1) to the 07-18 rx-B spare-port open/short blocks and the ambient load, with T_{NS} fixed, is in `data-analysis/scripts/marjum-2026-07/noisewave/` (README for the numbers). Smooth noise-wave parameters do not fit those data; allowing the noise waves a phase delay between P and the LNA predicts a held-out termination far better but leaves a residual well above the block-to-block repeatability. That residual can be absorbed either by a consistent error in the measured receiver reflection or by structure in T_{unc}, and the field data cannot tell which. A bench measurement of the receiver's reflection at P, through the same switch path, decides it; the study's README lists the measurements in order. Until then tier-2 temperatures do not replace tier 1.
- **Balun and coax.** Their available gain belongs in the comparison with any simulation. The coax did not survive; the balun can be measured.
- **RFI mask.** `flags@v3` (memo 006) removes most of the lines `flags@v2` left, but some aircraft-band and band-edge lines remain in the night's spectrum, and its RFI bits do not cover int32 wraps; mask negative samples as well.
- **Coverage.** rx-A 16:22–19:19 and the regime gap are NaN. Bridging them needs a model of the drift across the antenna raise, not a wider limit.

Revision history

- 2026-10-05: built on tcal v0003 and `flags@v3`; shows current results only.
- 2026-10-03: the contaminated calibration visits identified as box-air's 1.000 MHz comb EMI; the noise-wave open question points to the first tier-2 study.
- 2026-10-01: first version. Restates legacy MEMO-004 (noise-source drift, reproduced independently in §5; the in-band T_{NS} route in §7) and the calibration method of legacy MEMO-003. MEMO-003's receiver-temperature stability and uncertainty budget are not yet restated here, so that file stays until they are.

A Reproducing this memo

Run `python make_products.py` in this directory, with `EIGSEP_CAMPAIGN_ROOT` pointing at a campaign tree that holds the raw data, then `pdflatex memo-004.tex` twice. Tracked products are read from this checkout. The script checks the raw S11 files against `data/s11/manifest.csv`, records a digest of the correlator files it reads, and writes every figure in `figures/` and every table and quoted number in `tables/`. The tcal product is built by `data-analysis/scripts/marjum-2026-07/tcal/` (build, validate, bracket cost).

input	SHA-256 (first 16)
marjum-2026-07/derived/tcal/v0003/solutions.npz	7fc84f90ef8ff6ce
marjum-2026-07/data/s11/cal_materials/switch_sparams.npz	8b4418f07187c965
eigsep_cal/src/eigsep_cal/data/calibrated_internal_osl2026.npz	832ba837e98017f5
marjum-2026-07/data/s11/cal_materials/ vna_internal_osl_20260629T090825Z.npz	3a0e93ff130e4287
eigsep_data/hfss_beam_maps/bowtie_beam.npz	c4046f5e9332cd57
eigsep_sim/src/eigsep_sim/data/S11_eigsep_bowtie_v000.npz	a8abe8bc6ad9a9d8
data-analysis/notebooks/arp/marjum-2026-07/d5_yfactor.py	2471b6fab7128275
40 field S11 files (checked against manifest.csv)	all match
292 correlator files of the 16/17 Jul night (digest of SHA-256s)	a2f684c6cd25bd16
eigsep commit	90a0eb9
eigsep_cal commit	2c5cf99
eigsep_data commit	e941c51
data-analysis commit	0aa3364
eigsep_sim commit	8965eee
tcal v0003 built from data-analysis	0aa408c
generated	2026-10-05 20:30 UTC